PRE-RELEASE

eCH-1234 – Template Quarto Document

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Summary

Brief summary of the purpose of the document.

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Note

This document uses a gender-neutral formulation when referring to persons. This is based on the guidelines (German) of the Federal Chancellery. Depending on the situation, paired forms (citizens), gender-abstract forms (insured person), gender-neutral forms (insured person) or paraphrases without personal reference are used. The generic masculine (citizen) is not permitted. Full forms are used in continuous texts, i.e. in texts consisting of formulated sentences. Short forms can be used in abbreviated text passages, namely in tables. The short form is used with a slash but without an ellipsis (referent). Gender asterisks and similar spellings are not used.

1 Introduction

1.1 Status

In progress: Use is only permitted within the Technical Unit or the Expert Committee.

1.2 Scope of application

The information in this chapter should provide the reader with a brief overview of what this standard is intended for. Information about the following matters may be helpful here.

1.3 We can have sub-headings

And write some text.

1.3.1 Also sub-sub-headings

And write some more text. Maybe even with a pretty image.

When displaying diagrams, try to write them in Mermaid JS straight away; this makes changes in the future or translations straightforward.



Figure 1: Always add some text to describe what the image shows. If a figure has a caption, it appears in the list of figures. Only the first sentence of the caption is shown there.

1.4 Formatting

This chapter is a stress test: it contains the common Markdown and Quarto elements so that their rendering in the PDF and on the website can be checked.

1.4.1 Text styling

Text can be **bold**, *italic*, underlined, highlighted, IN SMALL CAPS or ~~struck through~~. Keyboard shortcuts such as `Ctrl-C` + `Ctrl-V` and inline code such as `schema:name` are set off visually.[1]

1.4.1.1 A fourth-level heading

Headings down to the fourth level are possible; they are numbered but do not appear in the table of contents.[2]

1.4.2 Lists and callouts

Unordered list:

- First item
- Second item
  - First sub-item
  - Second sub-item
    - First detail
    - Second detail
      - First note
      - Second note
- Third item

Ordered list:

1. First step
2. Second step
   1. First sub-step
   2. Second sub-step
      1. First detail
      2. Second detail
3. Third step

> **Note**
>
> A note draws attention to connections that are easily overlooked.

> **Warning**
>
> A warning points out sources of error, for example unchecked input data.

[1] On the website, footnotes appear in the right margin; in the PDF, at the bottom of the page.
[2] A second footnote with a link to ech.ch.

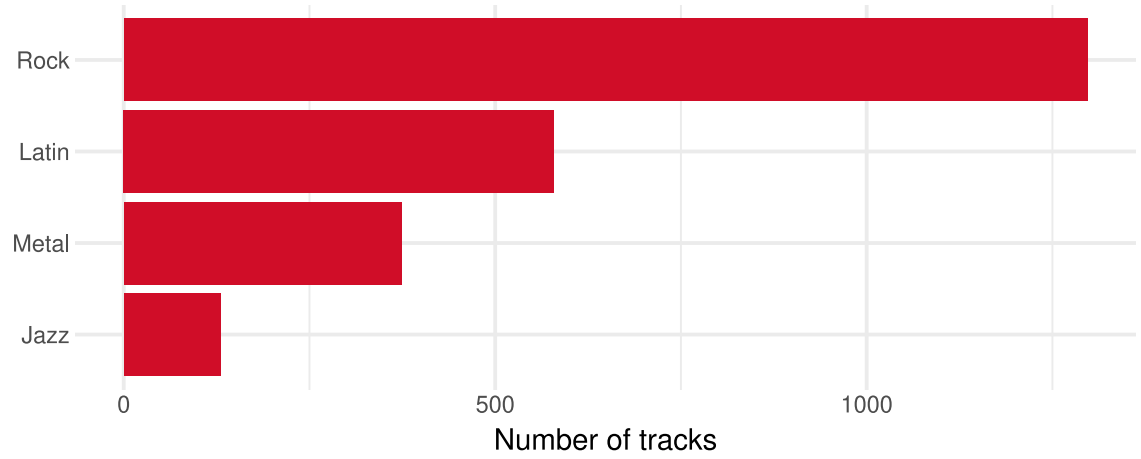


Figure 2: Figure produced with Python. Quarto runs the code when rendering and inserts the plot.

1.4.3 Code, tables and formulas

The following code block shows a SPARQL query:

```sparql
SELECT ?s ?p ?o
WHERE {
  ?s ?p ?o .
}
LIMIT 10
```

Figures can also be produced with Python code, as Figure 2 shows:

Table 1: Example table with three columns. If a table has a caption, it appears in the list of tables. Only the first sentence of the caption is shown there.

Column	Type	Remark
Name	String	Mandatory field
Quantity	Decimal number	with unit

A formula can appear in the text, for example $E = mc^2$, or as a standalone equation such as Equation 1:

$$\sum_{i=1}^{n} x_i = n\bar{x} \tag{1}$$

The following paragraph is deliberately nonsense and only tests the citation style (in running text without brackets with `@key`, otherwise in brackets with `[@key]`): according to eCH-0003 11.1.0, all postal addresses [eCH-0010 8.1.0] must be serialised in Turtle [Turtle], validated as a DCAT catalogue [DCAT; eCH-0200 3.0.1] with SHACL [SHACL] and published via SPARQL [SPARQL] as Linked Open Data [eCH-0205 1.0] on agricultural crops [eCH-0265 1.0.0] in JSON-LD [JSON-LD] and OWL [OWL 2], as the Semantic Web [Berners-Lee 2023] intends.

2 Usage notes

2.1 Namespaces

Table 2 lists the prefixes and namespaces used in this document. A prefix stands for a namespace IRI, the common beginning of the IRIs of a vocabulary. The Turtle specification describes how such *prefixed names* are resolved [Turtle].

Table 2: Namespaces used in eCH-1234 – Template Quarto Document.

Prefix	Namespace
:	https://agriculture.ld.admin.ch/eCH-1234/2/
codelist:	https://agriculture.ld.admin.ch/eCH-1234/2/codelist/
country:	https://ld.admin.ch/country/
currency:	http://qudt.org/vocab/currency/
media:	https://agriculture.ld.admin.ch/eCH-1234/2/mediatype/
rdf:	http://www.w3.org/1999/02/22-rdf-syntax-ns#
schema:	http://schema.org/
unit:	http://qudt.org/vocab/unit/
xsd:	http://www.w3.org/2001/XMLSchema#

3 Data Model

3.1 Genre

A musical category in the Chinook dataset.

3.1.1 Overview

Shape	:GenreShape
Target class	:Genre
Closed	Yes (only the listed properties are allowed)
Key figures	25 instances with 5 to 7 triples each (5.4 on average)
Data access	SPARQL service

3.1.2 Properties

Description	Details	Cardinality
Name (schema:name)	• Type: Text (xsd:string) • Pattern: `^[A-Z]` • Length: 2–50 characters	1..1
Part of (schema:partOf)	• Type: Genre	0..*
Alternate name (schema:alternateName): Translated or colloquial name of the genre, at most one per language.	• Type: Language-tagged text (rdf:langString) • Length: ≤ 50 characters • Languages: `de`, `fr`, `it`, `en` (one per language)	0..*

3.1.3 Rules (SPARQL)

3.1.3.1 No cyclic hierarchy

A genre cannot be a part of itself, either directly or transitively.

```
SELECT $this
WHERE {
    $this schema:partOf+ $this .
}
```

3.2 Invoice

A purchase receipt in the Chinook dataset. An invoice groups the items of one purchase. The invoice total is not only recorded as a value but also checked against the sum of its items; differences of more than five cents count as errors. Invoices dated before the year 2000 stem from a legacy system and are not permitted in this dataset.

3.2.1 Overview

Shape	:InvoiceShape
Target class	schema:Invoice

Closed	No (further properties are allowed)
Key figures	412 instances with 8 to 21 triples each (12.4 on average)
Data access	SPARQL service

3.2.2 Properties

Description	Details	Cardinality
Customer (schema:customer): An invoice must be linked to exactly one customer.	• Type: Person	1..1
Total payment due (schema:totalPaymentDue): An invoice must define a total payment due as a schema:QuantitativeValue.	• Type: Quantitative Value with the following restrictions: ‣ Unit Code (schema:unitCode): Required value currency:USD	1..1
Has part (schema:hasPart): An invoice must have at least one line item (OrderItem).	• Type: schema:OrderItem	1..*
Date created (schema:dateCreated)	• Type: Date (xsd:date) • Range: ≥ 2000-01-01 • Examples: 2001-01-01, 2024-05-21	0..*

3.2.3 Rules (SPARQL)

3.2.3.1 Invoice date

An invoice cannot be dated before the year 2000.

```
SELECT $this
WHERE {
    $this schema:dateCreated ?date .
    FILTER(?date < "2000-01-01"^^xsd:date)
}
```

3.2.3.2 Invoice total

Invoice total payment due must exactly match the sum of its lines (within 0.05 tolerance).

```
SELECT $this
WHERE {
    $this schema:totalPaymentDue/schema:value ?invoiceTotal .
    {
        SELECT $this (SUM(?price * ?qty) AS ?calculatedTotal)
        WHERE {
            $this schema:hasPart ?line .
            ?line schema:price/schema:value ?price .
            ?line schema:orderQuantity/schema:value ?qty .
        }
```

```
        GROUP BY $this
    }
    FILTER(ABS(?invoiceTotal - ?calculatedTotal) > 0.05)
}
```

3.3 Music album

A collection of tracks in the Chinook dataset.

3.3.1 Overview

Shape	:MusicAlbumShape
Target class	schema:MusicAlbum
Closed	No (further properties are allowed)
Key figures	347 instances with 5 triples each
Data access	SPARQL service

3.3.2 Properties

Description	Details	Cardinality
Name (schema:name): Every album must have a name.	• Type: Text (xsd:string) • Length: ≤ 200 characters	1..1
Artist (schema:byArtist): Person or music group who created the album.	• Type: Person or schema:MusicGroup	0..*

3.4 Music Recording

A single music track in the Chinook dataset.

3.4.1 Overview

Shape	:TrackShape
Target class	schema:MusicRecording
Closed	No (further properties are allowed)
Key figures	3503 instances with 10 to 15 triples each (11.1 on average)
Data access	SPARQL service

3.4.2 Properties

Description	Details	Cardinality
Name (schema:name): Every track must have a name.	• Type: Text (xsd:string) • Length: 1–200 characters	1..1
In Album (schema:inAlbum): A track can only belong to a valid schema:MusicAlbum.	• Type: Music album	0..*
Author (schema:author): The person or group who wrote the track.		0..*

Description	Details	Cardinality
Genre (schema:genre)	• Type: Genre	1..1
Media type (schema:encodingFormat): The file format of the track as a code of the media types code list.	• Type: schema:CategoryCode with the following selection: ▸ MPEG audio file (media:1) ▸ Protected AAC audio file (media:2) ▸ Protected MPEG-4 video file (media:3) ▸ Purchased AAC audio file (media:4) ▸ AAC audio file (media:5) See code list Media types	1..1
Duration (schema:duration): The duration of the track as a quantitative value in milliseconds with a value greater than zero.	• Type: Quantitative Value with the following restrictions: ▸ Unit Code (schema:unitCode): Required value unit:MilliSEC ▸ Value (schema:value): Range > 0	1..1
Content Size (schema:contentSize): The file size of the track as a quantitative value in bytes with a value greater than zero.	• Type: Quantitative Value with the following restrictions: ▸ Unit Code (schema:unitCode): Required value unit:BYTE ▸ Value (schema:value): Range > 0	1..1

3.5 Organisation

A company or organization in the Chinook dataset.

3.5.1 Overview

Shape	:OrganisationShape
Target class	schema:Organisation
Closed	No (further properties are allowed)
Data access	None (transactional data, not part of the published graph)

3.5.2 Properties

Description	Details	Cardinality
Name (schema:name): Every organisation must have a name.	• Type: Text (xsd:string) • Length: 2–100 characters	1..1

3.6 Person

Any person (employee, customer, or custom person) present in the dataset.

Persons are described uniformly regardless of their role: employees of the music store, customers and manually recorded persons share the same properties. The role follows from the relationships, for example `schema:worksFor` for employees or `schema:customer` on an invoice for customers.

Names are recorded as the person uses them; academic titles do not belong in the given name. Email addresses and birth dates are optional and rule violations are reported as warnings only.

3.6.1 Overview

Shape	:PersonShape
Target class	schema:Person
Closed	Yes (only the listed properties are allowed)
Key figures	383 instances with 5 to 10 triples each (5.6 on average)
Data access	SPARQL service

3.6.2 Properties

Description	Details	Cardinality
Given Name (schema:givenName): Every person must have a given name.	• Type: Text (xsd:string)	1..1
Family Name (schema:familyName): Every person must have a family name.	• Type: Text (xsd:string)	1..1
Email Address (schema:email): If an email is provided, it must follow a standard email format.	• Type: Text (xsd:string) • Pattern: `^.+@.+\..+$` (`i`) • Length: ≤ 254 characters • Severity: Warning	0..*
Birth date (schema:birthDate): A person should have a valid birth date.	• Type: Date (xsd:date) • Range: ≥ 1900-01-01, ≤ 2025-12-31 • Severity: Warning	0..*
Address (schema:address)	• Type: Postal address	0..*
Works for (schema:worksFor): An employee can report to another person.	• Type: Person or schema:Organization	0..*
Job title (schema:jobTitle)	• Type: Text (xsd:string) • Length: 2–100 characters	0..1
knows (schema:knows)	• Type: Person	0..*

3.6.3 Rules (SPARQL)

3.6.3.1 No self-reporting

An employee cannot report to themselves.

```
SELECT $this
WHERE {
    $this schema:worksFor $this .
}
```

3.7 Postal address

The postal address of a person.

Addresses follow the structure of schema.org rather than the Swiss address standard eCH-0010, since the dataset contains mostly foreign addresses. Postal codes are therefore only checked for a plausible format, not against a directory.

3.7.1 Overview

Shape	:PostalAddressShape
Target class	schema:PostalAddress
Closed	Yes (only the listed properties are allowed)
Key figures	67 instances with 6 to 8 triples each (7.5 on average)
Data access	SPARQL service

3.7.2 Properties

Description	Details	Cardinality
Street address (schema:streetAddress)	• Type: Text (xsd:string) • Length: 3–100 characters • Examples: Schwarzenburgstrasse 165	1..1
Postal code (schema:postalCode): Letters, digits, spaces and hyphens, two to ten characters.	• Type: Text (xsd:string) • Pattern: `^[A-Za-z0-9][A-Za-z0-9 -]{1,9}$` • Severity: Warning	0..1
Locality (schema:addressLocality): Every address needs exactly one locality.	• Type: Text (xsd:string) • Length: ≤ 100 characters	1..1
Region (schema:addressRegion): Canton, state or province where customary in the country.	• Type: Text (xsd:string) • Length: ≤ 100 characters	0..1
Country (schema:addressCountry): Country as an IRI from the country list of the Swiss Federal Archives on LINDAS (ISO 3166-1 alpha-3).	• Pattern: `^https://ld\.admin\.ch/country/[A-Z]{3}$` • Examples: country:CHE, country:FRA, country:ITA	1..1

3.8 Quantitative Value

A numerical value with an associated unit.

Quantitative values are never recorded as a bare number but always together with a unit from the units code list (IRIs of the QUDT vocabulary). This keeps it unambiguous whether a duration is meant in milliseconds or a file size in bytes. Negative values are not foreseen in this dataset.

3.8.1 Overview

Shape	:QuantitativeValueShape
Target class	schema:QuantitativeValue
Closed	Yes (only the listed properties are allowed)

Key figures 15401 instances with 3 triples each
Data access SPARQL service

3.8.2 Properties

Description	Details	Cardinality
Value (schema:value): A quantitative value must have exactly one numeric value.	• Type: Decimal number (xsd:decimal) or Integer (xsd:integer) • Range: ≥ 0	1..1
Unit Code (schema:unitCode): The unit as a code of the units code list (an IRI of the QUDT vocabulary).	• Type: schema:CategoryCode, see code list Units • Examples: currency:USD, unit:MilliSEC	1..1

4 Code lists

The following code lists define the permitted values of the corresponding properties of the data model (Section 3). Every code list is published on LINDAS as a `schema:CategoryCodeSet` and every value as a `schema:CategoryCode`.

4.1 Media types

The file formats of the tracks of the Chinook dataset. The code values are the identifiers of the table MediaType of the Chinook database.

Table 3: Values of the code list Media types.

IRI	Code	Name	Description
media:1	1	MPEG audio file (MP3)	Audio file in the MPEG-1 Audio Layer III format (MP3), without copy protection.
media:2	2	Protected AAC audio file (M4P)	AAC audio file with copy protection (DRM), as sold by the iTunes Store until 2009.
media:3	3	Protected MPEG-4 video file (M4V)	MPEG-4 video file with copy protection (DRM), such as purchased films and series.
media:4	4	Purchased AAC audio file (iTunes Plus)	AAC audio file purchased in the iTunes Store without copy protection (iTunes Plus).
media:5	5	AAC audio file (M4A)	AAC audio file without copy protection, e.g. imported from a CD.

4.2 Units

The units of the quantitative values of the dataset: duration in milliseconds, file size in bytes, number of items and prices in US dollars. The codes are IRIs of the QUDT vocabulary, the code values the UN/CEFACT common codes or the ISO 4217 currency code.

Table 4: Values of the code list Units.

IRI	Code	Name	Description
unit:BYTE	AD	Byte (B)	Eight bits; unit of the file size of the tracks.
unit:MilliSEC	C26	Millisecond (ms)	One thousandth of a second; unit of the duration of the tracks.
unit:NUM	EA	Each (#)	Counting unit for numbers of items, such as the ordered quantity of an invoice line.
currency:USD	USD	US dollar ($)	Currency of the United States (ISO 4217: USD); unit of the prices and invoice totals.

5 Data Retrieval

The master and reference data underlying this document are available as *Linked Data*.

The technological basis for this is the Resource Description Framework [RDF, RDF], a central standard of the World Wide Web Consortium (W3C) for modeling data structures on the web. In RDF, information is not represented in classic tables, but as interconnected graphs. Each statement consists of a so-called triple (subject, predicate, object). This structure enables a machine-readable, interoperable, and cross-system unambiguous description of resources and their relations to one another.

For the storage and publication of this RDF data, LINDAS (Linked Data Service) is used, the official Linked Data service of the Swiss Federal Administration. LINDAS functions as a so-called *triple store*, a specialized graph database optimized for the efficient storage and querying of RDF triples, which makes the data publicly available via a standardized interface.

The following chapter provides minimal instructions on how the data can be queried and retrieved from LINDAS.

```
BASE <https://agriculture.ld.admin.ch/eCH-1234/2/>
PREFIX schema: <http://schema.org/>
SELECT *
WHERE {
    ?genre a <Genre> ;
        schema:name ?name .
}
LIMIT 10
```

The underlying data itself is maintained on GitHub as Turtle files.

```
@base <https://agriculture.ld.admin.ch/eCH-1234/2/> .
@prefix genre: <https://agriculture.ld.admin.ch/eCH-1234/2/genre/> .
@prefix schema: <http://schema.org/> .

genre:1 a <Genre> ;
    schema:name "Rock" .

genre:2 a <Genre> ;
    schema:name "Jazz" .

genre:3 a <Genre> ;
    schema:name "Metal" ;
    schema:partOf genre:1 .
```

6 Safety considerations

Information about the explicitly relevant legal bases or a note that during the implementation the relevant legal bases must be observed.

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Annex A – References

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Annex B – Cooperation and Verification

Name	Organisation
Damian Oswald	Federal Office for Agriculture
Michael Schüpbach	Federal Office for Agriculture
Lea Stauber	Federal Office for Agriculture

Annex C – Abbreviations and Glossary

This glossary is also available in machine-readable form as a SKOS glossary [SKOS], on LINDAS and in the GitHub repository.

Table 5: Glossary of the eCH-1234 standard

Term	Description
Graph (Named graph)	A set of triples. A named graph is itself identified by an IRI and allows data in a triple store to be grouped. *Broader*: Resource Description Framework
Internationalized Resource Identifier (IRI)	Globally unique identifier of a resource. In RDF, subjects, predicates and most objects are identified by IRIs. *Broader*: Resource Description Framework
JSON for Linking Data (JSON-LD)	A serialization of RDF based on JSON, particularly suited to web interfaces. *Broader*: Serialization
Linked Data (Linked Open Data)	Data published according to the principles of the Semantic Web: identified by IRIs, described in RDF and linked to one another. *Related*: Linked Data Service, Resource Description Framework
Linked Data Service (LINDAS)	The official Linked Data service of the Swiss Federal Administration, acting as a triple store.
Literal	A concrete value in RDF, such as a string, a number or a date, optionally with a datatype or a language tag. *Broader*: Resource Description Framework
Namespace (Prefix)	Common beginning of a group of IRIs. In Turtle and SPARQL, a namespace is abbreviated by a prefix, for example `schema:` for `http://schema.org/`. *Related*: Internationalized Resource Identifier

Term	Description
Ontology	A formal model of a domain that describes classes, properties and their logical relationships in a way that allows machines to draw conclusions. *Broader*: Vocabulary *Related*: Web Ontology Language
Reasoning (Inference)	The automatic derivation of new statements from the existing data and the axioms of an ontology, for example by the HermiT reasoner.
Resource Description Framework (RDF)	A central standard of the World Wide Web Consortium (W3C) for modeling data structures on the Web. Information is represented as networked graphs rather than in traditional tables.
Serialization	Textual representation of an RDF graph in a defined format, used to store or exchange it. *Narrower*: JSON for Linking Data, Terse RDF Triple Language
Shape	A set of conditions that a class of resources or a property must fulfil, such as mandatory fields, datatypes and cardinalities. *Broader*: Shapes Constraint Language
Shapes Constraint Language (SHACL)	The W3C language for describing and validating the structure of RDF data. The data model of this standard is formulated in SHACL. *Narrower*: Shape
Simple Knowledge Organization System (SKOS)	A W3C vocabulary for thesauri, classifications and glossaries. This glossary is itself recorded in SKOS. *Related*: Vocabulary
SPARQL endpoint	A web interface through which SPARQL queries can be sent to a triple store. *Broader*: SPARQL Protocol and RDF Query Language
SPARQL Protocol and RDF Query Language (SPARQL)	The W3C query language for RDF data. SPARQL is used to read data from graphs, modify them and exchange them between systems. *Related*: Resource Description Framework
Terse RDF Triple Language (Turtle)	A compact, human-readable serialization for RDF. All RDF files of this standard are written in Turtle. *Broader*: Serialization
Triple	The basic structure of a statement in RDF, consisting of a subject, predicate, and object. *Broader*: Resource Description Framework
Triple store (Graph database)	A database specialised in storing and querying RDF triples. *Related*: Linked Data Service

Term	Description
Vocabulary	A collection of classes and properties with defined meaning that is reused to describe data, for example schema.org. *Narrower*: Ontology
Web Ontology Language (OWL)	The W3C language for formulating ontologies. It allows classes and properties to be defined with logical axioms. *Related*: Reasoning

Annex D – Changes in comparison to previous version

The most important changes of version 2.0.0 compared to the previous version 1.4.6:

- **Document template:** the PDF output now follows the eCH document template (title page with metadata table, header and footer, table of contents, annexes); the website adopts the same style in a light and a dark variant.
- **Metadata:** number, category, quality stage, version, status and further details of the standard are recorded once and inserted automatically into all language versions, the title page, the footer and Section 1.1.
- **Lists:** the table of figures and the table of tables (Annexes E and F) are generated automatically and linked to the figures and tables.
- **Presentation:** uniform styling of tables, captions and code in the PDF and on the website.

A complete list of all changes per version can be found in the releases on GitHub.

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